Red Hat OpenShift AI Self-Managed 3.4

Working with the model catalog

Working with the model catalog in Red Hat OpenShift AI Self-Managed

Last Updated: 2026-05-15

Red Hat OpenShift AI Self-Managed 3.4 Working with the model catalog

Working with the model catalog in Red Hat OpenShift AI Self-Managed

Abstract

Discover and evaluate generative AI models in the AI hub model catalog and select models to register, deploy, and customize.

Table of Contents

PREFACE

As a data scientist or AI engineer in OpenShift AI, you can discover and evaluate the generative AI models that are available in the AI hub model catalog. From the model catalog, you can select the models to register, deploy, and customize.

CHAPTER 1. OVERVIEW OF THE MODEL CATALOG AND MODEL REGISTRIES

The model catalog provides a curated library where data scientists and AI engineers can discover and evaluate the available generative AI (gen AI) models to find the best fit for their use cases.

A model registry acts as a central repository for administrators, data scientists, and AI engineers to register, version, and manage the lifecycle of AI models before configuring them for deployment. A model registry is a key component for AI model governance.

1.1. MODEL CATALOG

Data scientists and AI engineers can use the model catalog to discover and evaluate the gen AI models that are available and ready for their organization to register, deploy, and customize.

The model catalog provides models from different providers that you can search, discover, and evaluate before you register models in a model registry and deploy them to a model serving runtime. Third-party gen AI models are benchmarked by Red Hat for performance and quality by using open-source evaluation datasets. You can compare performance metrics for specific hardware configurations and determine the most suitable option for deployment.

OpenShift AI provides a default model catalog that includes models from different source providers. OpenShift AI administrators can manage and govern model catalog sources to control which models are available to data scientists and AI engineers. For more information, see Manage and govern model catalog sources.

For more information about how data scientists and AI engineers discover and evaluate models in the catalog, see Working with the model catalog.

1.2. MODEL REGISTRY

A model registry is an important component in the lifecycle of an artificial intelligence/machine learning (AI/ML) model, and is a vital part of any machine learning operations (MLOps) platform or workflow. A model registry acts as a central repository, storing metadata related to machine learning models from development to deployment. This metadata ranges from high-level information like the deployment environment and project, to specific details like training hyperparameters, performance metrics, and deployment events.

A model registry acts as a bridge between model experimentation and serving, offering a secure, collaborative metadata store interface for stakeholders in the ML lifecycle. Model registries provide a structured and organized way to store, share, version, deploy, and track models.

OpenShift AI administrators can create model registries in OpenShift AI and grant model registry access to data scientists and AI engineers. For more information, see Managing model registries.

Data scientists and AI engineers with access to a model registry can use it to store, share, version, deploy, and track models. For more information, see Working with model registries.

CHAPTER 2. DISCOVERING AND EVALUATING MODELS IN THE MODEL CATALOG

You can discover and evaluate the available gen AI models in the model catalog to find the best fit for your use cases. You can select from available model categories, search by text, and filter by labels.

For validated models, you can view performance benchmark data for specific hardware configurations to evaluate and compare options for deployment.

Prerequisites

- You are logged in to the Red Hat OpenShift AI dashboard.
- The model registry component is enabled in your OpenShift AI deployment. For more information, see Enabling the model registry component.

Procedure

1. From the OpenShift AI dashboard, click **AI hub** → **Models** → **Catalog**.
2. The **Catalog** page provides a high-level view of available models, including the model category, name, description, and labels such as task, license, and provider. You can also view performance benchmarks for validated models from third parties.
3. In the menu bar, select from the available model categories:
   - **All models**: All models available in the model catalog.
   - **Red Hat AI models**: Models provided and supported by Red Hat.
   - **Red Hat AI validated models**: Third-party models benchmarked by Red Hat for performance and quality by using open-source evaluation datasets.
   - **Other models**: Custom third-party and community models configured by your administrator that do not have any catalog source labels. This category is only displayed if there are catalog sources without labels.
     Otherwise, custom models with labels configured by your administrator are displayed in a category with the same name as the label set for the custom catalog source.

   > **NOTE**
   >
   > On the **s390x** architecture, only the **granite-3.3-8b-instruct** model is supported. Its model card is available from the **All models** category in the menu bar of the model catalog.

4. You can use the search bar to find a model in the catalog. Enter text to search by model name, description, or provider.
5. You can use the filter menu to search and select filters by the following labels:
   - **Task**: For example, **Text-generation**.
   - **Provider**: For example, **Meta**.
   - **License**: For example, **Apache 2.0**.

- **Language**: For example, **Japanese**.
- **Tensor type**: For example, **BF16**.

6. Click the name of a model to view the model details page. This page displays the model description and the **Model card** information supplied by the model provider. This includes details such as the model's intended use and potential limitations, training parameters and datasets, and evaluation results.

7. For validated models, click the **Performance Insights** tab to view performance benchmark data to compare performance metrics for specific hardware configurations and to determine the most suitable options for deployment.
   You can filter the performance data by the following options:

   - **Scenario**: Select a workload type to view performance for different input and output token lengths, for example, **Chatbot**.
   - **Latency**: Set your maximum acceptable latency. Hardware configurations with response times above this value are not displayed. You can select a specific metric in the list:
     - **E2E** (end-to-end request latency): The time taken from submitting the request to receiving the final response.
     - **TTFT** (time to first token): The time that the user must wait before seeing output from the model.
     - **TPS** (tokens per second): The total number of tokens that are output per second.
     - **ITL** (inter-token latency): The average time taken between consecutive tokens.
       You can also select a percentile value, for example, **P90**.

       Use the slider to set the maximum acceptable latency value in milliseconds, and click **Apply filter**.
   - **Max RPS**: Set the minimum number of requests per second that displayed models must support. Hardware configurations that perform below this value are not displayed.
     Use the slider to set the maximum requests per second value, and click **Apply filter**.
   - **Hardware**: Select one or more hardware types from the list, for example, **H200**.
     You can click **Clear all filters** to reset your filters and try again.

8. For categories with more than 10 models, you can click **Load more models** to scroll and view additional models available in the catalog. Repeat this step until all models are loaded.

Verification

- For all models, you can view the information about a selected model on the model details page.
- For validated models, you can view the benchmark information about a selected model on **Performance Insights** tab.

CHAPTER 3. ABOUT THE MODEL PERFORMANCE VIEW

The **Model performance view** displays only validated models in the model catalog, and enables you to filter models based on advanced workload and hardware constraints. The recommended hardware configurations, throughput (output tokens), and latency metrics are derived from Pareto-optimal filtering, which provides non-dominated options across all performance dimensions.

When you select a validated model card in the catalog, the active performance filter constraints are transferred to the **Performance Insights** tab on the model details page. You can immediately compare compression variants and hardware options against the specified constraints in the catalog. This streamlines the process of moving from initial discovery to selecting the optimal model and hardware combination for a specific production workload.

Modifying the performance filter settings changes all the displayed model cards and their content to align with the applied constraints. Disabling and enabling the **Model performance view** resets the performance filter to the default settings.

When you enable the **Model performance view** view, models are sorted by lowest to highest latency and the model cards show validated models only and performance data with labels instead of description and labels. In addition, the time to first token, inter-token, and end-to-end latency metrics change according to the filters that you select in the workload and performance constraints.

CHAPTER 4. VIEWING PERFORMANCE DATA FOR VALIDATED MODELS

You can use the **Model performance view** to display only validated models and to filter models by using advanced workload and hardware metrics. You can specify a workload type (Chatbot, RAG, Long RAG, or Code Fixing), service level objectives for latency metrics at different percentiles, and a maximum request per second threshold. This ensures that only models and hardware configurations that meet the specified deployment requirements are displayed.

Prerequisites

- You are logged in to the Red Hat OpenShift AI dashboard.

Procedure

1. In the OpenShift AI dashboard, click **AI hub → Models → Catalog**.
2. Click the **Model performance view** toggle in the left filter pane of the catalog.
   This enables the performance filters, displays model benchmark data, excludes unvalidated models, and sorts models from lowest to highest latency.
3. Change the default performance filter settings that apply to all validated models. The default settings are as follows:
   - **Scenario**: Chatbot
   - **Latency**: Time to first token for 90th percentile
   - **Max RPS**: 1
   - **Hardware**: none
4. To apply the active performance filters to a specific model, click the model name on a model card. The performance filters transfer automatically to the **Performance Insights** tab for the model.
5. To transfer the filter changes made on the **Performance Insights** tab back to the model catalog, click **Catalog** in the breadcrumb navigation.

   > **NOTE**
   >
   > This transfer occurs only if you enable the **Model performance view** before accessing the model details. Returning to the model catalog by using the **Models** link in the left navigation panel deactivates the **Model performance view**.

6. On the **Performance Insights** tab for a model, you can also click **Customize columns** to add or remove columns from the **Performance Insights** table. Customizing the columns does not require the performance filters to be active.

Verification

- The model catalog displays only validated models, sorted by latency from lowest to highest.

- Each model card displays performance metrics (hardware, replicas, and time to first token latency) instead of the model description.

CHAPTER 5. ABOUT TENSOR TYPES AND MODEL VARIANTS

Each tensor type is a compression variant of the same base model, which means that the same model can have multiple variants with different trade-offs in performance. All models include tensor type and size on their details page in the model catalog.

A validated model can have multiple variants, each with a different tensor type. For example, the Llama-3-8B model can have the following variants:

- Llama-3-8B-FP16 (16-bit floating point) - full precision, highest quality, heavier resource requirements.
- Llama-3-8B-INT8 (8-bit integer) - quantized to 8 bits, good balance between quality and efficiency.
- Llama-3-8B-INT4 (4-bit integer) - quantized to 4 bits, lightest, lower accuracy.

Each model variant has its own performance artifacts: recommended hardware, throughput, and latency can vary significantly between them.

Pareto-optimal filtering takes model variants into account when recommending configurations in the catalog. For example, an INT4 variant can achieve the same requests per second with less hardware than an FP16 variant.

CHAPTER 6. FILTERING AND ASSESSING MODELS BY TENSOR TYPE

You can use a model’s **Performance Insights** tab to compare model variants based on tensor types and to decide which variant best fits your deployment requirements such as hardware cost and response quality.

> **NOTE**
>
> The **Performance Insights** tab includes a **Model variants by tensor type** comparison card only for validated models that have variants.

Prerequisites

- You are logged in to the Red Hat OpenShift AI dashboard.

Procedure

1. In the OpenShift AI dashboard, click **AI hub → Models → Catalog**.
2. In the filter pane on the left, scroll down to the filter options for **Tensor type**.
3. Select one or more tensor types, for example, FP16 (16-bit floating point) and BF16 (brain floating point).
   The model catalog is filtered to show only models that have variants matching the selected tensor types.

   > **NOTE**
   >
   > The **Tensor type** filter is always available and works regardless of whether the **Model performance view** toggle is enabled.

4. Click a validated model to display its model details.
5. Click the **Performance Insights** tab.
   This tab contains a **Model variants by tensor type** comparison card for compression variants based on tensor types.
6. Click a model compression variant to visually assess the trade-off between compression and quality for the selected model in the metrics displayed on the **Performance Insights** tab.

   > **NOTE**
   >
   > If you enabled the **Model performance view** and set active filters in the catalog, those filters are automatically carried over to the **Performance Insights** tab.

Verification

- The catalog displays only models with variants matching the selected tensor types.
- The **Performance Insights** tab for a selected model displays the **Model variants by tensor type** comparison card.

CHAPTER 7. REGISTERING A MODEL FROM THE MODEL CATALOG

As a data scientist or AI engineer, you can register models directly from the model catalog and create the first version of the new model.

Prerequisites

- You are logged in to Red Hat OpenShift AI.
- You have access to an available model registry in your deployment.

Procedure

1. From the OpenShift AI dashboard, click **AI hub → Models → Catalog**.
2. The **Catalog** page provides a high-level view of available models, including the model category, name, description, and labels such as task, license, and provider.
3. You can use the search bar to search by model name, description, or provider.
4. You can use the filter menu to search and select filters by task, provider, or license.
5. Click the name of a model to view the model details page.
6. Click **Register model**.
7. From the **Model registry** drop-down list, select the model registry that you want to register the model in.
8. In the **Model details** section, configure details to apply to all versions of the model:
    a. Optional: In the **Model name** field, update the name of the model.
    b. Optional: In the **Model description** field, update the description of the model.
9. In the **Version details** section, enter details to apply to the first version of the model:
    a. In the **Version name** field, enter a name for the model version.
    b. Optional: In the **Version description** field, enter a description for the first version of the model.
    c. In the **Source model format** field, enter the name of the model format, for example, **ONNX**.
    d. In the **Source model format version** field, enter the version of the model format.
10. In the **Model location** section, the URI of the model is displayed.
11. Click **Register model**.

    NOTE

    Registering models from the model catalog is not supported on the **s390x** architecture.

Verification

- The new model details and version are displayed on the **Overview** tab on the model details page.
- The new model and version are displayed on the **Model registry** page.

CHAPTER 8. DEPLOYING A MODEL FROM THE MODEL CATALOG

You can deploy models directly from the model catalog.

> **NOTE**
>
> OpenShift AI model serving deployments use the global cluster pull secret to pull models in OCI-compliant ModelCar format from the catalog.
>
> For more information about using pull secrets in OpenShift, see Updating the global cluster pull secret in the OpenShift documentation.

Prerequisites

- You have completed the prerequisites in Deploying models.
- The model registry component is enabled in your OpenShift AI deployment. For more information, see Enabling the model registry component.

Procedure

1. From the OpenShift AI dashboard, click **AI hub → Models → Catalog**.
2. The **Catalog** page provides a high-level view of available models, including the model category, name, description, and labels such as task, license, and provider.
3. You can use the search bar to search by model name, description, or provider.
4. You can use the filter menu to search and select filters by task, provider, or license.
5. Click the name of a model to view the model details page.
6. Click **Deploy model** to display the **Deploy a model** wizard.
7. On the **Model details** page, in the **Model type** field, you can select **Generative AI model** or **Predictive model**. The default model type in the catalog is generative.

   > **NOTE**
   >
   > For models in the catalog, the **Model details** page displays read-only information in the **Model location** and **URI** fields.

8. In the **Model deployment** page, configure the deployment as follows:
   a. From the **Project** list, select the project in which to deploy your model.
   b. In the **Model deployment name** field, enter a unique name for your model deployment. This field is autofilled with the model name by default.
      This is the name of the inference service created when the model is deployed.
   c. Optional: Click **Edit resource name**, and enter a specific name in the **Resource name** field. By default, the resource name matches the name of the model deployment.

IMPORTANT

Resource names are what your resources are labeled as in OpenShift. Your resource name cannot exceed 253 characters, must consist of lowercase alphanumeric characters or -, and must start and end with an alphanumeric character. You cannot edit resource names after creation.

The resource name must not match the name of any other model deployment resource in your OpenShift cluster.

d. In the **Description** field, enter a description of your deployment.

e. From the **Hardware profile** list, select a hardware profile. Models provided in the catalog use the **default-profile**.

f. Optional: To modify the default resource allocation, click **Customize resource requests and limits** and enter new values for the CPU and memory requests and limits.

g. In the **Serving runtime** field, select one of the following options:

- **Auto-select the best runtime for your model based on model type, model format, and hardware profile**
  The system analyzes the selected model framework and your available hardware profiles to recommend a serving runtime.

- **Select from a list of serving runtimes, including custom ones**
  Select this option to manually choose a runtime from the list of global and project-scoped serving runtime templates.

  For more information about how the system determines the best runtime and administrator overrides, see Automatic selection of serving runtimes.

h. Optional: For predictive AI models only, you can select a framework from the **Model framework (name - version)** list. This field is not displayed for generative AI models.

i. In the **Number of model server replicas to deploy**field, specify a value.

j. Click **Next**.

9. On the **Advanced settings** page, configure the following options:

- Select the **Add as AI asset endpoint**checkbox if you want to add your gen AI model endpoint to the **Gen AI studio → AI asset endpoints** page.
  - In the **Use case** field, enter the types of tasks that your model performs, such as chat, multimodal, or natural language processing.

  NOTE

  You must add your model as an AI asset endpoint to test your model on the **Gen AI studio → playground** page.

- To require token authentication for inference requests to the deployed model, select **Require token authentication**.

- In the **Service account name** field, enter the service account name that the token will be generated for.
  - To add an additional service account, click **Add a service account** and enter another service account name.
- In the **Configuration parameters** section:
  - Select **Add custom runtime arguments**, and then enter arguments in the text field.
  - Select **Add custom runtime environment variables**, and then click **Add variable** to enter custom variables in the text field.
- In the **Deployment strategy** section, select one of the following options:
  - **Rolling update**: Existing inference service pods are terminated *after* new pods are started. This ensures zero downtime and continuous availability. This is the default option.
  - **Recreate**: Existing inference service pods are terminated *before* new pods are started. This saves resources but guarantees a period of downtime.

10. On the **Review** page, review the settings that you have selected before deploying the model.
11. Click **Deploy model**.

NOTE

Advanced settings are not supported on the **s390x** architecture.

Verification

- The model deployment is displayed in the following places in the dashboard:
  - The **AI hub → Deployments** page.
  - The **Latest deployments** section of the model details page.
  - The **Deployments** tab for the model version.

Additional resources

- Deploying models on the model serving platform .